

Active Disturbance Rejection Based Iterative Learning Control for Variable Air Volume Central Air-Conditioning System

Shiying Lu¹, Wei Ai^{2*}, Xiangyang Li³

Key Laboratory of Autonomous Systems and Networked Control, Ministry of Education,
School of Automation Science and Engineering, South China University of Technology, Guangzhou 510641, China
1 E-mail: :ceeinglu@163.com, 2 E-mail: aiwei@scut.edu.cn, 3 E-mail:xyangli@scut.edu.cn

Abstract: The Variable Air Volume (VAV) Central Air-Conditioning system is a complicated system with non-linearity, large-time delay and strong inertia, thus it is difficult to design an effective controller. Iterative Learning Control (ILC) takes good effect in controlled process with repeatability and periodicity, but it cannot cope with uncertain disturbance explicitly. A creative algorithm, Active Disturbance Rejection based Iterative Learning Control (ADR-Based ILC), is proposed to improve ILC's performance in VAV control system. ADR-Based ILC compensates the disturbance explicitly caused by ambient temperature, heat from people and machines, and makes it higher control precision and higher energy-efficiency. An accurate model of VAV system is built in TRNSYS platform, and ADR-Based ILC is proved to be more effective than fuzzy PID and ILC.

Key Words: Variable Air Volume, Central Air-Conditioning system, Active Disturbance Rejection based Iterative Learning Control, TRNSYS

1 Introduction

The Variable Air Volume (VAV) air-conditioning system changes the air volume transferred into the air-conditioning region as to adjust the contemporary change of corresponding region heat loads. Due to its flexible, energy saving and comfortable advantages, VAV air-conditioning system is becoming more and more popular among the large air-conditioning systems nowadays. However, the features of VAV air-conditioning system (such as multi-loop, nonlinear and strong coupling) make the design and operation of the VAV system complicated and difficult. The whole VAV system is a huge and complex entirety which contains several components including building, chiller, pump and cooling tower, couples of water and air circulation. Furthermore, the non-linearity, time-delay, strong inertia and tight coupling add difficulty to the control of the complicated system. It also brings huge challenges to the control of the whole Central Air-Conditioning (CAC) system. Therefore, investigation of VAV system control techniques is important and practical significance [1-5].

Classical PID controllers are still widely used in the VAV systems. They fail to meet the need of energy efficiency and comfort level. To get better energy efficiency and enhance the operation performance, many intelligent algorithms are proposed to optimize the PID controller. Liu Kun designed an adaptive fuzzy PID controller and applied it into the VAV air conditioning system, which obtained a satisfactory result [2]. A decoupling control theory based on state feedback was proposed concerning the strong coupling of VAV system in [3], which avoided high order of decoupling control compensator and made it practically feasible. Zuojun Liu

proposed a flexible iterative learning control (ILC) in air-conditioning system, taking advantages of historical data and operation experience in [4]. Ref. [6] studies PID control method and Active Disturbance Rejection Control (ADRC) technique of VAV air-conditioning system.

To some extent, the CAC system operates periodically. The effective control strategies for the periodic process include ILC and repetitive learning control. When the ILC has zero initial value, it can achieve full tracking under controlled conditions. For a periodic process system, both the internal disturbance and the external disturbance vary periodically. ILC solves the problem of periodic uncertainties via tuning the control input iteration, but it is indirect and implicit. Ref. [7, 8] proposed a data-driven adaptive ILC method, which compensated the uncertainties by iterative domain dynamic linearization. It combines the DDC method with ILC for the first time. Considering the definition of DDC control and its dynamic linearization strategy [9, 10], to some extent we think that ADRC can also be regarded as a DDC method. As the core of ADRC, Extended State Observer (ESO) is a very effective tool to estimate the uncertainties in time domain [11-13]. How about extending the ESO's compensation range into iterative domain?

From this point of view, a new ILC method based on ADRC is proposed [14]. Here, an iterative ESO (LESO) strategy is presented to estimate the uncertainties in iteration domain explicitly via I/O data during the iteration process. Thus, the uncertainties such as ambient temperature and occupant load in CAC will be compensated by LESO, and TD arranges the transition process to solve the zero initial value problem of ILC. By this iterative compensation way, the unknown disturbance from the last cycle becomes the known disturbance of the next cycle. Combined traditional ILC with ADRC, this method has more effective disturbance rejection ability and convergence rate.

The rest of this paper is organized as follows. The ADR-Based ILC is explicitly introduced in Section 2. The

*This work is supported by Guangdong Science and Technology Plan Project under Grant 2013A011402003; Natural Science Foundation of Guangdong Province of China (2016A030313505); National Natural Science Foundation of China (61773170).

*Corresponding author' email: aiwei@scut.edu.cn.

model of VAV system is constructed based TRNSYS platform in Section 3. The control result of ADR-Based ILC and the comparison with other algorithms are showed in Section 4. At the end of this paper, the conclusions of the research are highlighted in Section 5.

2 Active Disturbance Rejection Based Iterative Learning Control Algorithm

2.1 Linear Extended State Observer

The core of ADRC-Based ILC Algorithm is iterative domain ESO, which can estimate the uncertainties from iterations. We discuss the traditional time-domain ESO first.

Consider this nonlinear time-varying system

$$\begin{cases} \dot{x}_i(t, k) = x_{i+1}(t, k); & i = 1, \dots, n-1 \\ \dot{x}_n(t, k) = f(t, x) + b(t, x)u(t, k); & t \in [0, T] \end{cases} \quad (1)$$

It is a 2-D system, where x_i is the system state, r_i is the desired trajectory, t and k are time variable and iteration variable respectively. Here u is the control signal, n presents system order, $f(t, x)$ and $b(t, x)$ are unknown uncertainties of the system.

Denote $x^T = [x_1, \dots, x_n]$ and $r^T = [r_1, \dots, r_n]$.

The control objective lies in how to find an appropriate u which will force the state x to trace the desired trajectory r .

Some assumptions are given as follows [11-13].

Assumption 1: The desired trajectory: $r(t)$ and $\dot{r}_n(t)$ are bounded.

Assumption 2: $f(t, x)$ and $b(t, x)$ are bounded and locally Lipschitz continuous, where $\frac{\partial f}{\partial t}$ is bounded.

Assumption 3: System (1) is with the identical initial condition.

Assumption 4: $b_0(t, x)$ is a crude estimate of $b(t, x)$. Both $b(t, x)$ and $b_0(t, x)$ have the same sign.

Then the error equation of (1) is

$$\begin{cases} \dot{e}_i(t, k) = e_{i+1}(t, k); & i = 1, \dots, n-1 \\ \dot{e}_n(t, k) = f_{be}(t, x) + b_0 u(t, k) - \dot{r}_n \end{cases} \quad (2)$$

where

$$f_{be}(t, x) = f(t, x) + (b - b_0)u(t, k) \quad (3)$$

$f_{be}(t, x)$ can be estimated very well by time domain ESO. Thus we have the following linear ESO about the time domain:

$$\begin{cases} \hat{x}_i(t, k) = \hat{x}_{i+1} - \frac{\alpha_{n+1-i}}{\varepsilon^i} (\hat{x}_1 - x_1); & i = 1, \dots, n-1 \\ \hat{x}_n(t, k) = \hat{x}_{n+1} - \frac{\alpha_1}{\varepsilon^n} (\hat{x}_1 - x_1) + b_0 u(t, k) \\ \hat{x}_{n+1}(t, k) = -\frac{\alpha_0}{\varepsilon^{n+1}} (\hat{x}_1 - x_1) \end{cases} \quad (4)$$

For linear ESO (4), there are two conclusions in the condition described in [4]:

1. The linear ESO's estimation error converges uniformly to limit zero for every positive constant T_ε

$$\lim_{\varepsilon \rightarrow 0} |x_i(t, k) - \hat{x}_i(t, k)| = 0 \quad t \in [T_\varepsilon, \infty) \quad (5)$$

2. The limit superior satisfies

$$\overline{\lim}_{t \rightarrow \infty} |x_i(t, k) - \hat{x}_i(t, k)| \leq O(\varepsilon^{(n+2-i)}) \quad (6)$$

where $i = 1, \dots, n+1$, and $x_{n+1} = f_{be}(t, x)$ is the extended state of system (1).

As for the iteration problem, time domain ESO in (4) doesn't work efficiently, because ESO's estimation accuracy will not be enhanced through iteration process.

2.2 Linear Iterative Extended State Observer

Take the following coordinate transformation for linear ESO in (4)

$$\begin{cases} z_1(t, k) = \hat{x}_1(t, k) \\ z_i(t, k) = \hat{x}_i - \sum_{j=1}^{i-1} \frac{\alpha_{n+1-j}}{\varepsilon^j} (z_{i-j} - x_{i-j}) \end{cases} \quad (7)$$

where $i = 2, \dots, n+1$

First we define

$$\begin{cases} z_0(t, k) = \int_0^t z_1(\tau, k) d\tau \\ x_0(t, k) = \int_0^t x_1(\tau, k) d\tau \end{cases} \quad (8)$$

Therefore

$$\begin{cases} \dot{z}_i(t, k) = z_{i+1}; & i = 0, \dots, n-1 \\ \dot{z}_n(t, k) = -\sum_{j=0}^n \frac{\alpha_j}{\varepsilon^{n+1-j}} \delta_j(t, k) + b_0 u \end{cases} \quad (9)$$

where

$$\begin{cases} \delta_j(t, k) = x_j(t, k) - z_j(t, k) \\ \delta_j(0, k) = 0 \quad j = 0, \dots, n \end{cases} \quad (10)$$

From (7) we obtain

$$x_i - z_i = x_i - \hat{x}_i + \sum_{j=1}^{i-1} \frac{\alpha_{n+1-j}}{\varepsilon^j} (x_{i-j} - z_{i-j}) \quad (11)$$

So we get the following inequality with absolute value

$$|x_i - z_i| \leq |x_i - \hat{x}_i| + \left| \sum_{j=1}^{i-1} \frac{\alpha_{n+1-j}}{\varepsilon^j} (x_{i-j} - z_{i-j}) \right| \quad (12)$$

Here, both z_i and $\hat{x}_i$ can be regarded as the accurate estimation of x_i . Then we can write the error equation of system (9) as shown below

$$\begin{cases} \dot{\delta}_i(t, k) = \delta_{i+1}(t, k); & i = 0, \dots, n-1 \\ \dot{\delta}_n(t, k) = f_{b\delta}(t, x) + b_0 u(t, k) - \dot{x}_n \end{cases} \quad (13)$$

where

$$f_{b\delta}(t, x) = -\sum_{j=0}^n \frac{\alpha_j}{\varepsilon^{n+1-j}} \delta_j \quad (14)$$

By observing system (1) and system (9), error equation (2) and (13), they have the same structure just with different variable notations and can be regarded as so-called dual systems [14]. It gives the iterative ESO algorithm to estimate f_{be} skillfully by analogy.

$$\begin{cases} w_r(t, 0) = 0, & k \in \mathbb{Z}^+ \\ w_r(t, k) = w_r(t, k-1) + \sum_{j=0}^n \frac{\alpha_j}{\varepsilon^{n+1-j}} e_j(t, k) \end{cases} \quad (15)$$

where

$$e_0(t, k) = \int_0^t e_1(\tau, k) d\tau \quad (16)$$

2.3 Control Law

Now define the iterative-domain compensation $\sigma(t, k)$ as

$$\sigma(t, k) = \frac{\varepsilon}{\alpha_n} \sum_{j=0}^n \frac{\alpha_j}{\varepsilon^{n+1-j}} e_j(t, k) \quad (17)$$

Thus,

$$\begin{cases} w_r(t, 0) = \sigma(t, 0) = 0, & k \in \mathbb{Z}^+ \\ w_r(t, k) = w_r(t, k-1) + \frac{\alpha_n}{\varepsilon} \sigma(t, k) \end{cases} \quad (18)$$

where $w_r(t, k)$ is the estimate of $f_{be}(t, x)$. Taking the derivative of $\sigma(t, k)$ with respect to t gives

$$\dot{\sigma}(t, k) = \frac{\varepsilon}{\alpha_n} \sum_{j=0}^{n-1} \frac{\alpha_j}{\varepsilon^{n+1-j}} \dot{e}_j(t, k) + \dot{e}_n(t, k) \quad (19)$$

Combining (2) and (19) yields

$$\dot{\sigma}(t, k) = \frac{\varepsilon}{\alpha_n} \sum_{j=0}^{n-1} \frac{\alpha_j}{\varepsilon^{n+1-j}} e_{j+1}(t, k) + f_{be}(t, x) + b_0 u(t, k) - \dot{r}_n \quad (20)$$

Design the control law as

$$u(t, k) = b_0^{-1} (\dot{r}_n - w_x(t, k) - \beta \sigma(t, k)) \quad (21)$$

where $\beta \geq 0$ is feedback gain and $w_x(t, k)$ is defined by

$$w_x(t, k) = w_r(t, k) + \frac{\varepsilon}{\alpha_n} \sum_{j=0}^{n-1} \frac{\alpha_j}{\varepsilon^{n+1-j}} e_{j+1}(t, k) \quad (22)$$

In [14] it is proved that the iterative law (18) and the control law (21) guarantee a wonderful system tracking performance. In the next we will apply this method in VAV control system.

3 Variable Air Volume System

3.1 Variable Air Volume System Construction

While taking the cooling mode into account, the VAV system includes building, chillers, pumps, cooling towers, cooling coils, fans, etc. These components make up a complicated CAC system that contains several air and water circulation, which could interact with each other, showed in Figure 1. Thus it is difficult to control each variable respectively due to the strong coupling[15]. What's more, the effect of hydromechanics adds the system with characteristics including non-linearity, long-time delay and strong inertia.

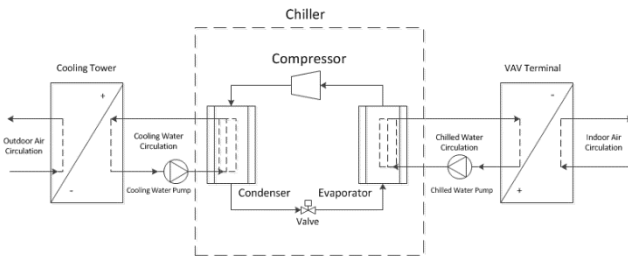


Fig. 1: Working process of a VAV CAC system

The end system mainly includes rooms and VAV fans. Here consider a teaching building as the plant. The building is a complicated system where temperature couples in classrooms, time-lag effect is too serious to neglect and external disturbance such as heat caused by people and machines is changeable. The above uncertainties outer

disturbance, coupled with inner disturbance, can be viewed as total disturbance. The object of ADR-Based ILC is to estimate and cancel total disturbance and accelerate error convergence to 0.

3.2 Mathematical Model

According to the thermo dynamic equation of the system [16]

$$C_1 \frac{dt_n}{dt} = Q - c\rho G(t_n - t_s) \quad (23)$$

where C_1 is the heat capacity of the room, c is the specific heat, ρ is density of air and the supply air volume is G , t_n and t_s are indoor temperature and supply air temperature.

Define t_n and t_s as

$$t_n = t_{n0} + \Delta t_n \quad (24)$$

$$t_s = t_{s0} + \Delta t_s \quad (25)$$

where t_n and t_s are indoor temperature and supply air temperature at the operating point.

Combining (23), (24) and (25) yields

$$\frac{dt_n}{dt} = \frac{Q - c\rho G(\Delta t_n - \Delta t_s)}{C_1} + \frac{c\rho(t_{s0} - t_{n0})}{C_1} G \quad (26)$$

Taking t_n as state variable x , G as control signal u

$$\dot{x} = f(x) + bu \quad (27)$$

Equation (27) complies with the general system form of ADR-Based-ILC proposed above. The total disturbance $f(x)$, containing thermal load of the room, is uncertain but it can be estimated and compensated by IESO. To some extent, thermal load changes periodically because of the regular curriculums and the performance of the controller improves as the times of iteration increase.

The 1-order ADR-Based IL Controller is designed as follow

$$e_1(t, k) = t_n - sp \quad (28)$$

$$e_0(t, k) = \int_0^t e_1(\tau, k) d\tau \quad (29)$$

$$\sigma(t, k) = \frac{\varepsilon}{\alpha_1} \left(\frac{\alpha_0}{\varepsilon^2} e_0(t, k) + \frac{\alpha_1}{\varepsilon} e_1(t, k) \right) \quad (30)$$

$w_r(t, k)$ can be considered as the estimate of uncertainty $f_{be}(t, k)$

$$w_r(t, k) = w_r(t, k-1) + \frac{\alpha_1}{\varepsilon} \sigma(t, k), k \in \mathbb{Z}^+ \quad (31)$$

$$w_r(t, 0) = \sigma(t, 0) = 0 \quad (32)$$

$$w_x(t, k) = w_r(t, k) + \frac{\varepsilon}{\alpha_1} \cdot \frac{\alpha_0}{\varepsilon^2} e_1(t, k) \quad (33)$$

The control law

$$u(t, k) = 1/b_0 (-\dot{r}_2 + w_x(t, k) + \beta \sigma(t, k)) \quad (34)$$

Choose the parameters of (30) and (34) as

$\omega_0 = 0.06$, $\alpha_0 = \omega_0^2$, $\alpha_1 = 2\omega_0$, $\varepsilon = 0.05$, $b_0 = 15$ where ω_0 is the bandwidth of the system.

3.3 Trnsys Platform Structure

The model of building is divided into 9 rooms and simulates the effect of weather, shading, ventilation and other heat disturbance, shown in Figure 2.

The model of VAV central air conditioner is shown in Figure 3. It settles 3 chillers to supply cooling air because of the heavy load of the teaching building. One of the chilled water circulation and cooling water circulation are highlighted in blue and red, respectively.

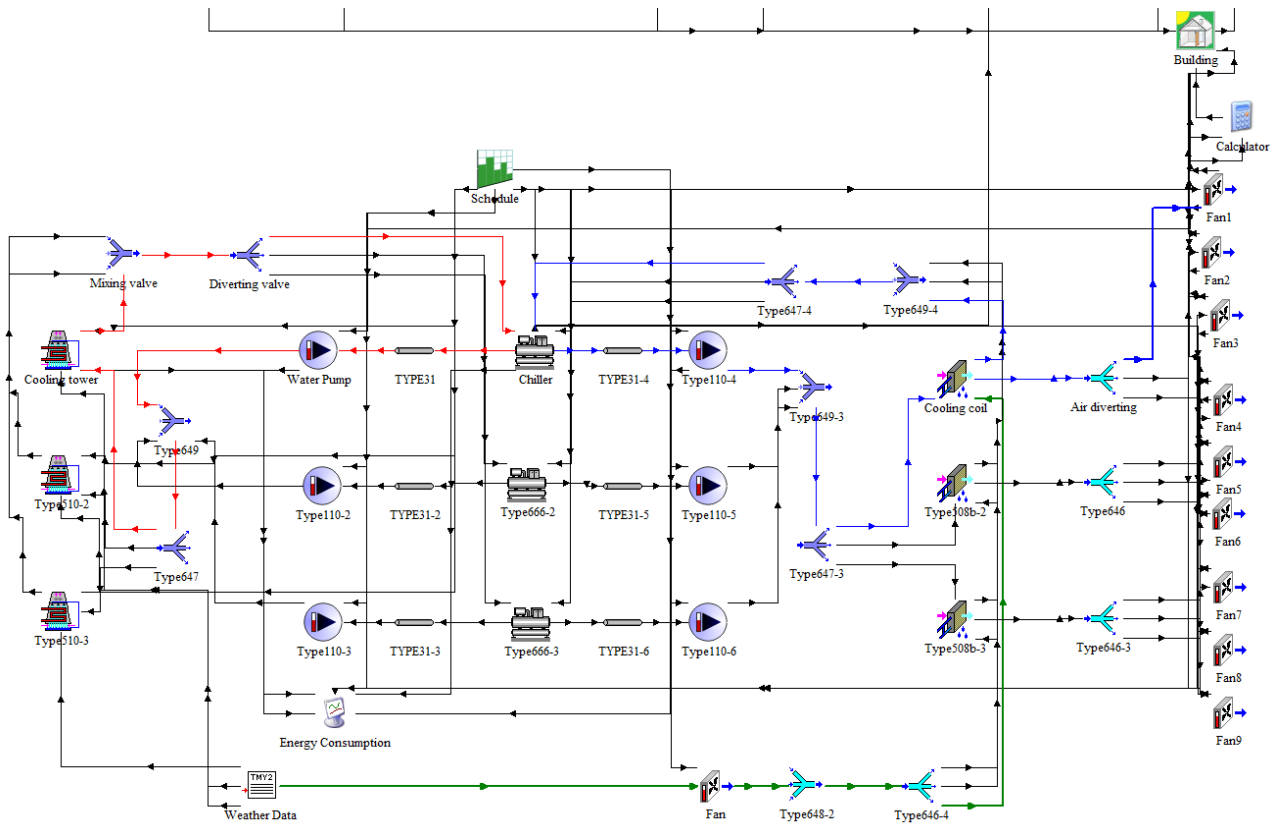


Fig. 3: The structure of VAV central air-conditioner in Trnsys platform

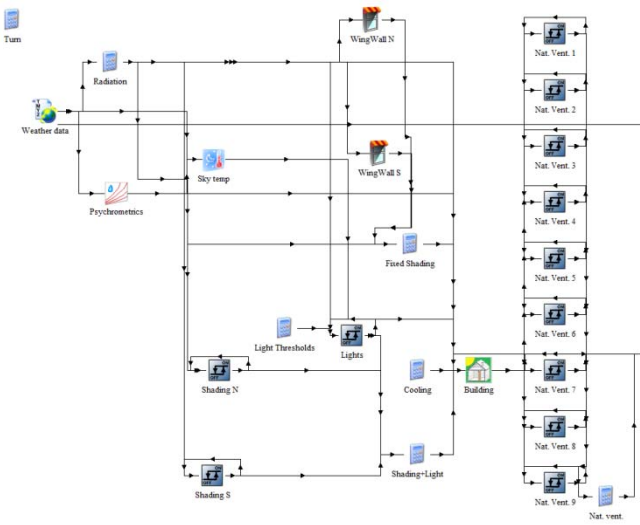


Fig 2.The structure and composition of building

4 Control Performance

The set point of room temperature is 26 Celsius. Nonlinear, time-varying, large disturbances are added to the system. The temperature under the control of ADR-Based ILC is showed in Figure 4, from which we can see the temperature converges to the reference signal quickly. Although overshoots exit in the first 3 days, they can be eliminated with the increase of iteration number. On the 7th day, the errors are within 0.1 Celsius.

The result of ADR-Based ILC, ILC, PID is tested within a week, showed in Figure 5. The 'COP' is given to identify the working time of VAV system. Notably, the steady-state error of ILC is unsatisfactory because ILC is ill-equipped to handle uncertainties and nonrepeating disturbances. PID control shows more oscillation and larger overshoots compared with ADR-Based ILC, showed in Figure 6.

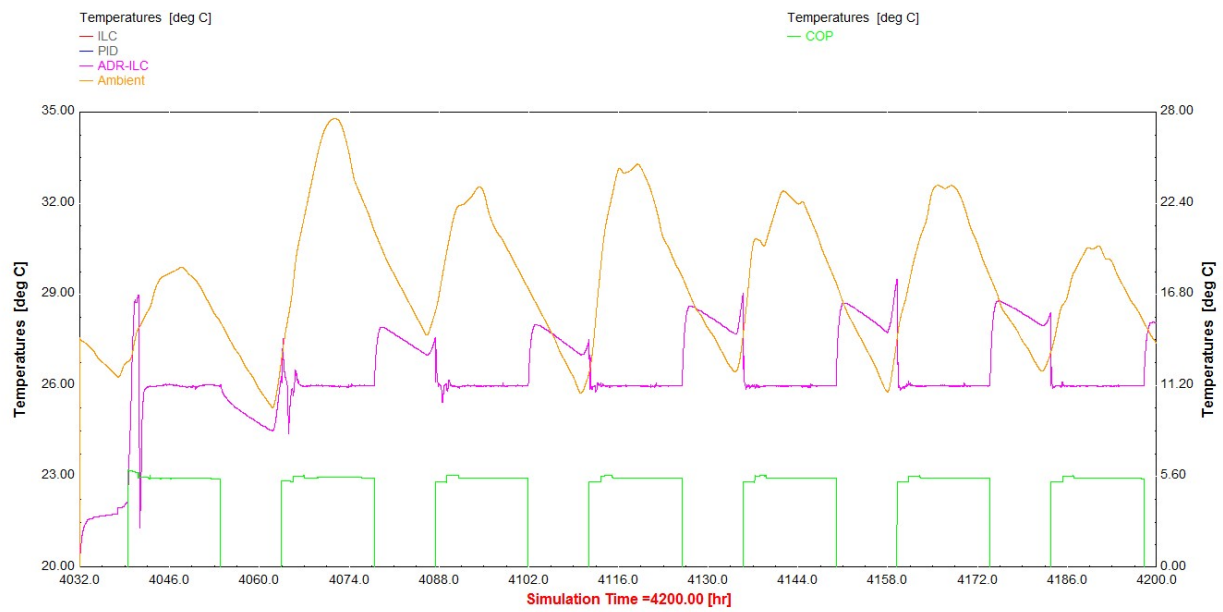


Fig. 4: Temperature under the control of ADR-Based ILC with strong disturbance

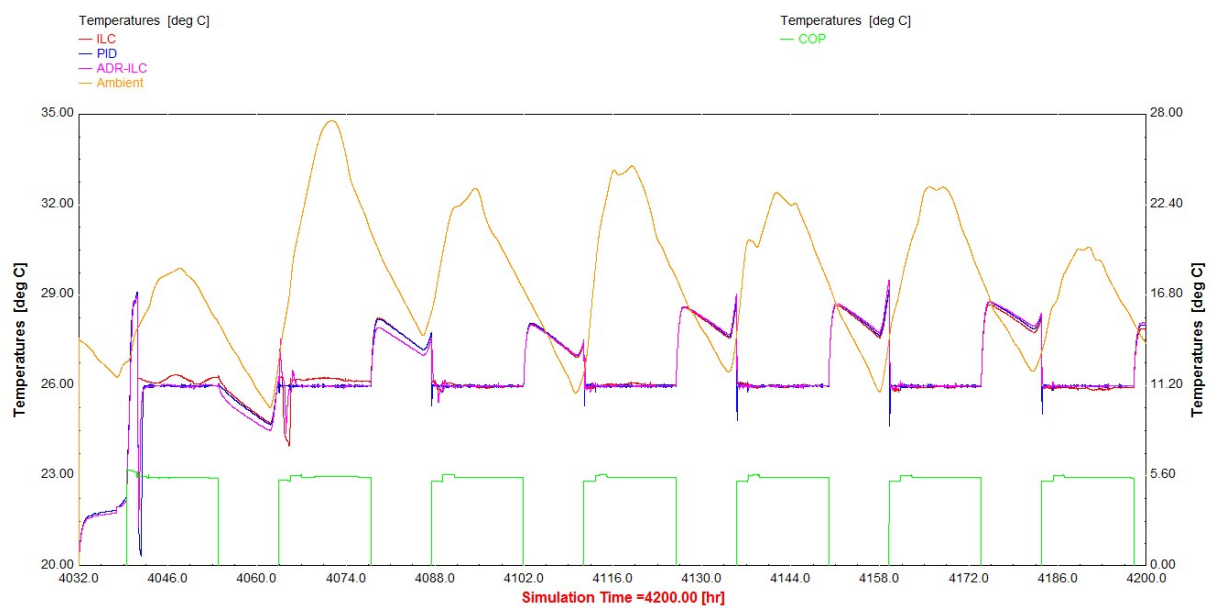


Fig. 5: Performance of ADR-Based ILC, ILC and PID

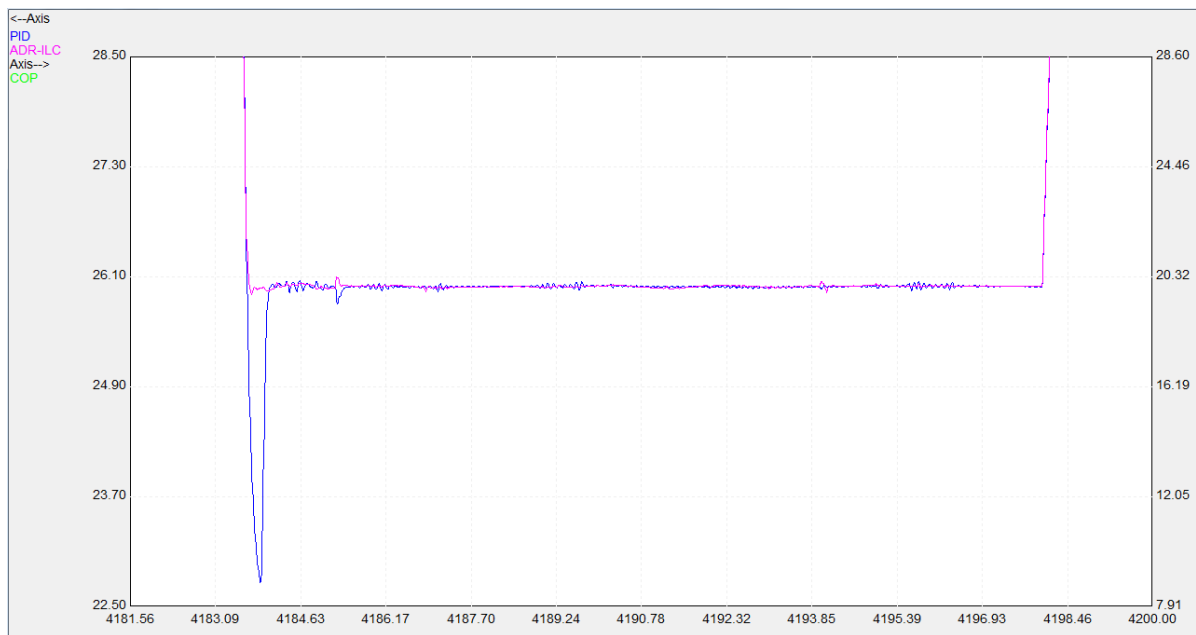


Fig. 6: Temperature curves of ADR-Based ILC and PID

5 Conclusions

The VAV air-conditioning systems have the characteristics of nonlinearity, large-time delay and strong coupling. ADRC effectively copes with the inner and outer disturbance of the VAV systems by compensating for it. The coupling, influence of weather and people can be classified as uncertain disturbance. ILC takes effect in repetitive and periodic controlled process by improving tracking accuracy as the increase of iteration times. The ADR-Based ILC algorithm, combining the advantages of ILC and ADRC, largely improves the control effect of the VAV system.

References

- [1] X. Zhang, R. Wang and J. Xing, Optimization control method of VAV air conditioning system, in *Proceedings of 12th World Congress on Intelligent Control and Automation*, 2016: 1105–1110.
- [2] L. Kun, and W. Dong, Fuzzy adaptive PID control for VAV air-condition system, in *Proceedings of 29th Chinese Control and Decision Conference*, 2017: 4354–4358.
- [3] Y. Huang, and X. Yang, Decoupling control of a Variable-Air-Volume Air-Conditioning system based on state feedback, in *Proceedings of 2nd International Symposium on Intelligent Information Technology Application*, 2008: 589–592.
- [4] Z. Liu, C. Cao, P. Yang, and Y. Zhou, Energy saving control in air conditioning system based on flexible Iterative Learning Control, in *Proceedings of 2nd International Symposium on Intelligent Information Technology Application*, 2008: 560–564.
- [5] D. Wei, and X. Liu, Research on multi-zone VAV air conditioning system modeling, in *Proceedings of the 10th World Congress on Intelligent Control and Automation*, 2012: 2968–2972.
- [6] Y. Sun, The Research on Variable Air Volume control system based on ADRC, Zhejiang University, 2008.
- [7] C. K. Yin, J. X. Xu, and Z. S. Hou, An ILC scheme for a class of nonlinear systems with time-varying parameters subject to second-order internal model, in *Proceedings of 28th Chinese Control and Decision Conference*, 2009: 452–457.
- [8] R. H. Chi, Z. S. Hou, and J. X. Xu, Adaptive ILC for a class of discrete-time systems with iteration-varying trajectory and random initial condition, *Automatic*, 44(8): 2207–2213, 2008.
- [9] Z. S. Hou, and Z. Wang, From model-based control to data-driven control: Survey, classification and perspective, *Information Sciences*, 235: 3–35, 2013.
- [10] Z. S. Hou, and J. Xu. On Data-driven control theory: the state of the art and perspective, *Acta Automatica Sinica*, 35(6):650–667, 2009.
- [11] J. Han. A class of extended state observers for uncertain systems, *Control and Decision*, 10(1): 85–88, 1995.
- [12] Q. Zheng, and Z. Q. Gao, Active disturbance rejection control: between the formulation in time and the understanding in frequency, *Control Theory and Technology*, 14(3): 250–259, 2016.
- [13] Y. Huang, W. C. Xue, and C. Z. Zhao, Active disturbance rejection control: methodology and theoretical analysis, *Journal of Systems Science & Mathematical Sciences*, 31(9): 1111–1129, 2011.
- [14] X. Y. Li, S. P. Tian, and A. Wei, Active disturbance rejection based iterative learning control. In *Proceedings of the 28th Chinese Control and Decision Conference*, 2016: 6645–6650.
- [15] Borresen B. A, Thermal room models for control analysis, *ASHRAE Transaction*, 87(2), 1981.
- [16] Tashtoush Bourhan, Molhim M, and Al-Rousan M. Dynamic model of an HVAC system for control analysis, *Energy*, 30: 1729–1745, 2005.